Week 42: Fair results from a bag

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Problems and routes

Problem	K–1	2–3	4–5	route
Every color pair, and which disappear without blue	P1			first
A shape rule with equal chances	P2			first
Count the shapes on the sixteen marked pairs	P3			first
The same rule on other bags	P4			return
Shapes as often as possible	P5			return
Six-pair stories	P6			return
Invent a fair rule and try it		P1		first
Check the rule on all sixteen labeled pairs		P2		first
The rule on other bags		P3		return
Which bag gives shapes more often?		P4		return
Changing bags between draws		P5		return
Can a rule promise a shape within six pairs?		P6		return
A fair rule with a shape on every pair?		P7		return
Design a fair color-only rule			P1	first
Check the rule on the labeled pairs			P2	first
Every fixed bag with both colors			P3	return
A rule that never skips			P4	return
One bag first, another second			P5	return
Returning only after the pair			P6	return
What fairness promises			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P2; Grades 4–5 P1–P2. **Return visits.** K–1 P4–P6; Grades 2–3 P3–P7; Grades 4–5 P3–P7.

Page numbers. A key that names a student page counts the pages of the full band packet. The route packets keep the problem numbers but number their own pages; each route-packet page is this page of the full packet:

Route packet	its pages = full-packet pages
K–1 first	pp. 1–3 = pp. 1–3
K–1 return	p. 1 = p. 1; pp. 2–4 = pp. 4–6
Grades 2–3 first	pp. 1–2 = pp. 1–2
Grades 2–3 return	p. 1 = p. 1; pp. 2–3 = pp. 3–4
Grades 4–5 first	p. 1 = p. 1
Grades 4–5 return	pp. 1–4 = pp. 1–4

Key, one entry per problem

Every color pair, and which disappear without blue

K–1 P1 · first route

Idea. Three red and one blue make four ordered color pairs, RR, RB, BR, BB; without blue only RR is left.

All four color pairs are RR, RB, BR, BB. Replacement makes BB possible even with only one blue counter. Taking the blue counter out leaves only RR: RB, BR, and BB disappear. A child may choose these draws deliberately to show possibility.

A shape rule with equal chances

K–1 P2 · first route

Idea. RB and BR have the same chance, so RB gives a square and BR a circle (or the reverse) is fair.

Use the rule stated above or its shape-swapped version. The six random outputs vary; even no outputs is possible. Do not judge the rule from a balanced or unbalanced sample. If a child proposes a different rule, keep it as a conjecture until the picture check in Problem 3.

Count the shapes on the sixteen marked pairs

K–1 P3 · first route

Idea. Of the 16 equally likely marked pairs, 3 are RB and 3 are BR, 9 are RR and 1 is BB; only RB against BR splits evenly.

RR has nine marked pictures; RB and BR have three each; BB has one. The two fair rules are exactly: (i) BR square, RB circle, RR/BB no shape; (ii) BR circle, RB square, RR/BB no shape. To see completeness, a shape receiving all nine RR pictures cannot be balanced by the seven other pictures, so RR is skipped. Among the remaining classes, BB cannot be balanced without breaking a three-picture class. Thus BB is skipped and RB/BR must go to different shapes. Adult-supported matching is enough; a written exhaustion is not required.

The same rule on other bags

K–1 P4 · return route

Idea. Mixed pairs RB and BR always have equal chances, so the rule stays fair on any bag with both colors; one-color bags give no shape. The marked-pair pictures show 3 and 3 of 16 for R, B1, B2, B3 and 4 and 4 for R1, R2, B1, B2.

One red/three blue gives each shape in 3 of 16 marked pairs (the R, B1, B2, B3 pictures on page 4). Two red/two blue gives each shape in 4 of 16 (the R1, R2, B1, B2 pictures). In both bags the shapes have equal positive chances. Four red or four blue gives no shape at all, so neither can give each shape.

Shapes as often as possible

K–1 P5 · return route

Idea. Under the mixed-pair rule, two red and two blue give shapes most often (8 of 16 marked pairs); no four-counter bag gives a shape on every pair.

Choose two red and two blue: the R1, R2, B1, B2 pictures on page 4 give a shape in 8 of 16. For red counts 0, 1, 2, 3, 4, the numbers of mixed labeled pairs are 0, 6, 8, 6, 0 out of 16. Hence 2-and-2 is optimal. No four-counter bag gives a shape on every pair under this rule: repeating the identity of the first counter is always possible and makes a same-color pair. A changed rule could: with two red and two blue, first color red gives a square and blue a circle is fair and never skips (8 and 8 of 16). That is why the problem says to keep your rule.

Six-pair stories

K–1 P6 · return route

Idea. Any number of random pairs can all be RR or BB, so no number of pairs forces a shape.

Examples of six-pair stories: **no shapes:** RR, RR, RR, RR, RR, RR. **Exactly one square, no circles:** BR, RR, RR, RR, RR, RR. **Exactly three circles, no squares:** RB, RB, RB, RR, RR, RR. Any ordering with the required outputs works. Six random pairs do not always give a shape: the first story has positive probability.

Invent a fair rule and try it

Grades 2–3 P1 · first route

Idea. Six trials cannot settle fairness; equal chances must come from the equally likely pairs.

Invent BR $\to$ square, RB $\to$ circle, RR/BB $\to$ no shape, or swap the shapes. Six trials cannot settle fairness.

Check the rule on all sixteen labeled pairs

Grades 2–3 P2 · first route

Idea. RB and BR each fill 3 of the 16 boxes, so the mixed-pair rule is exactly fair.

With labeled counters R1, R2, R3, B, all 16 ordered pairs are equally likely. In the grid below, rows are first draws and columns second draws.

R1: RR, RR, RR, RB. R2: RR, RR, RR, RB. R3: RR, RR, RR, RB. B: BR, BR, BR, BB.

The three RB cells and three BR cells have equal total weight. Pair (Ri,B) with (B,Ri). There are three outcomes for either shape and ten skipped outcomes; among the six kept outcomes each shape occupies half.

The rule on other bags

Grades 2–3 P3 · return route

Idea. For any bag with both colors, RB and BR have the same count of labeled pairs.

Both bags work. Grid heads: R1, R2, B1, B2 and R, B1, B2, B3. Two red/two blue gives four labeled RB and four BR outcomes out of 16. One red/three blue gives three RB and three BR outcomes. Reversing a pair pairs the outputs exactly, regardless of the observed random run.

Which bag gives shapes more often?

Grades 2–3 P4 · return route

Idea. Shapes come out on 6 of 16 pairs from three red and one blue, and on 8 of 16 from two and two.

The two-red bag produces a shape in 8 of 16 pairs; the three-red bag in 6 of 16. Choose the two-red bag. Both bags are fair when a shape comes out.

Changing bags between draws

Grades 2–3 P5 · return route

Idea. With different bags RB and BR can have different chances: 9 against 1 of 16 for RRRB then RBBB.

No. Grid rows R1, R2, R3, B; columns R, B1, B2, B3. Use the printed first bag **three red, one blue** and second bag **one red, three blue**. The exact counts are RR 3, RB 9, BR 1, BB 3 among 16 equally likely cross-bag identity pairs. For the key rule, circle has 9 ways and square 1; conditional on a shape, their chances are 9/10 and 1/10. Reversal no longer gives an equally likely outcome.

Can a rule promise a shape within six pairs?

Grades 2–3 P6 · return route

Idea. Six RR pairs are possible, so no promise holds.

No. RR repeated six times is a legal no-shape story. With replacement and the original bag its chance is $(9/16)^6$, which is positive. Children need only exhibit the legal story, not compute its chance. Hint: ask for a six-pair story that settles it.

A fair rule with a shape on every pair?

Grades 2–3 P7 · return route

Idea. From three red and one blue, a fair rule can never give a shape on every pair: the weights 9, 3, 3 and 1 (of 16) cannot be split into two groups of 8.

No color-only deterministic rule can be both fair and always produce a shape. All nine RR labeled pairs receive the same shape, already more than half of the 16 outcomes. The other shape cannot catch up even if it receives all seven remaining outcomes.

Design a fair color-only rule

Grades 4–5 P1 · first route

Idea. RB against BR is the only fair color-only rule for three red and one blue.

Use BR $\to$ square, RB $\to$ circle, RR/BB $\to$ no shape, or interchange the shapes.

Check the rule on the labeled pairs

Grades 4–5 P2 · first route

Idea. RB and BR each come from 3 of the 16 labeled pairs.

The four color classes have labeled sizes 9, 3, 3, 1. Reversal pairs each (Ri,B) with (B,Ri), giving equal shape probabilities 3/16 and conditional probabilities 1/2. Other rules fail Problem 2's check: RR $\to$ square, the rest $\to$ circle gives 9 against 7.

Every fixed bag with both colors

Grades 4–5 P3 · return route

Idea. For r red and b blue drawn with return, RB and BR each have chance $rb/(r+b)^2$: von Neumann's rule works for every fixed bag with both colors.

Both bags work: five red/two blue has 10 RB and 10 BR outcomes among 49; three red/four blue has 12 and 12 among 49. In general a fixed bag with r red and b blue has rb outcomes of each mixed order among $(r+b)^2$ equally likely identity pairs. The bag contents need not be known, but both r and b must be positive and the draws must follow the stated mechanism.

A rule that never skips

Grades 4–5 P4 · return route

Idea. A fair color-only rule for three red and one blue must skip at least 10 of the 16 labeled pairs.

Always giving a shape is impossible: the RR class alone has weight 9 > 8. If skipping is allowed, RR must still be skipped because all other classes total only 7. The remaining weights are 3, 3, 1. Their only equal, nonempty, disjoint subset sums are the two 3s. Therefore BB must also be skipped. The minimum is 10 skipped labeled pairs, attained by the two usual rules. This claim is restricted to deterministic color-only rules, not rules using identity marks or extra randomness.

One bag first, another second

Grades 4–5 P5 · return route

Idea. With different bags the rule is fair exactly when the two bags have the same red fraction.

First case: first bag 3R/1B, second 1R/3B gives RB 9/16 and BR 1/16, so unfair; a different rule, RR $\rightarrow$ square and BB $\rightarrow$ circle, is fair there, 3/16 each. **Second:** 3R/1B then 6R/2B gives RB and BR each 6/32 = 3/16, so fair; equal color proportions matter, not equal bag size. **Third:** 2R/2B then 1R/1B gives each mixed order 2/8 = 1/4, so fair. In general the equality p(1-q) = (1-p)q reduces to p = q; positive output additionally requires 0 < p < 1.

Returning only after the pair

Grades 4–5 P6 · return route

Idea. Without return inside a pair, RB and BR still have equal counts (3 and 3 of 12 ordered labeled pairs).

Yes. Without replacement there are 4 $\times$ 3 = 12 equally likely ordered pairs of distinct identities. RR has six, RB three, BR three, BB none. Reverse each kept pair to match its weight. Each shape has probability 3/12, and conditional on output it has chance 1/2. The within-pair draws are dependent, so the independent-draw product proof is inapplicable; the reversal proof still works. Another fair rule never skips here: both red $\rightarrow$ square, otherwise circle, 6 of 12 each.

What fairness promises

Grades 4–5 P7 · return route

Idea. Fairness is about chances, not about results in six pairs: neither equal counts nor any shape at all is guaranteed.

Neither claim follows. Six RR pairs produce no shape; BR, RR, RR, RR, RR, RR produces one square and no circles. Each story has positive probability under the original mechanism. Fairness says the possible outcomes have equal chances when a shape is produced; it does not force equal realized totals or a finite waiting bound.

Facilitator checkpoint: keep three claims separate - a legal example shows possibility, a finite trial suggests a conjecture, and equal-weight exhaustive counting or a bijection proves the exact probability claim.